STAT405 Final Report

Ruohan Sun, Haoyu You

April 2026

Public repository: https://github.com/HaoyuYou/Bayesian-Statistics-Project.git
Dataset link: https://www.kaggle.com/competitions/house-prices-advanced-regression-techniques

1 Introduction

House price prediction is a natural real-world Bayesian inference problem because sale prices are influenced by both observable house characteristics and latent neighborhood-level heterogeneity. We study the Ames housing dataset and model the log sale price using Bayesian regression. Our goal is, first, to construct a simple baseline model with interpretable predictors and, second, to propose a more flexible hierarchical extension that accounts for neighborhood-level variation.

A key challenge in this dataset is balancing interpretability and model flexibility. A naive model may capture the main linear trends but ignore structured differences across neighborhoods. A hierarchical model can partially pool neighborhood-level information and therefore provide a more realistic representation of the data-generating process.

2 Related work

De Cock (2011) introduced the Ames housing dataset as a richer alternative to the Boston housing dataset for regression analysis. Because it includes many structural housing characteristics, it is a useful benchmark for house price modeling.

House price analysis is often framed as a hedonic regression problem, where sale price is explained by property features and location. Previous studies have also shown that housing markets contain substantial local heterogeneity, so hierarchical or spatially varying models can improve interpretation and realism (Francke, Vos, & Janssen, 2004; Wheeler & Calder, 2014).

Our project differs from standard predictive analyses in two main ways. First, we focus on Bayesian inference and uncertainty quantification rather than only point prediction. Second, we compare two posterior computation methods,

Hamiltonian Monte Carlo and variational inference, and evaluate them using posterior summaries, diagnostics, and posterior predictive checks. This allows us to study not only house price modeling, but also the reliability of posterior approximation methods in a real-data Bayesian workflow. As noted by Stan Development Team (2024), variational inference can be computationally efficient, but it must be carefully validated before being trusted.

3 Data and preprocessing

We use the house price training data with sale price as the response. After preprocessing and removing rows with remaining missing values using `na.omit()`, the sample size is reduced from 1460 to 1126 observations. The response variable is transformed to

$$y_i = \log(\text{SalePrice}_i),$$

because the original sale price distribution is strongly right-skewed, while the log-transformed response is much closer to symmetric.

The preprocessing pipeline includes:

- explicit recoding of structural missingness such as `NoBsmt`, `NoGarage`, and `NoFireplace`,
- conversion of ordered quality variables to numeric scores,
- consolidation of sparse categorical levels,
- creation of derived variables such as `YearSinceRemodel`.

To construct a parsimonious baseline model, we examined the transformed response against candidate predictors and used a combination of correlation screening, collinearity checks, ANOVA-based exploratory screening, and interpretability. The final baseline predictors are

`OverallQual`, log(`GrLivArea`), `GarageCars`, `TotalBsmtSF`, `YearSinceRemodel`.

These variables were chosen because they show strong association with log sale price and retain a clear practical interpretation.

4 Model 1: Naive Bayesian regression

Our first model is a Gaussian regression on the log sale price:

$$y_i \sim \mathcal{N}(\mu_i, \sigma^2),$$

with linear predictor

$$\begin{aligned}\mu_i = \beta_0 &+ \beta_1 \text{OverallQual}_i + \beta_2 \text{LogGrLivArea}_i \\ &+ \beta_3 \text{GarageCars}_i + \beta_4 \text{TotalBsmtSF}_i + \beta_5 \text{YearSinceRemodel}_i.\end{aligned}$$

The prior specification is

$$\beta_0 \sim \mathcal{N}(8.8, 0.05^2),$$

$$\beta_1 \sim \mathcal{N}(0.10, 0.03^2), \quad \beta_2 \sim \mathcal{N}(0.25, 0.08^2), \quad \beta_3 \sim \mathcal{N}(0.06, 0.03^2),$$

$$\beta_4 \sim \mathcal{N}(0.00010, 0.00003^2), \quad \beta_5 \sim \mathcal{N}(-0.003, 0.004^2),$$

$$\sigma \sim \text{half-Normal}(0, 0.10^2).$$

These priors were chosen to be weakly informative but still centered around plausible magnitudes given the scale of the transformed response and the exploratory plots. Prior predictive checks showed that the implied curves and predictive densities were broadly compatible with the observed range of the data, so the priors were considered reasonable before posterior computation.

The Stan implementation of Model 1 is provided in the Appendix.

5 Model 2: Hierarchical varying-intercept and varying-slope model

To extend the baseline model, we introduce neighborhood-level heterogeneity through a hierarchical varying-intercept and varying-slope structure. Let $j[i]$ denote the neighborhood of house i. Then

$$y_i \sim \mathcal{N}(\mu_i, \sigma^2),$$

with

$$\begin{aligned}\mu_i = \beta_0 + \alpha_{j[i]} + (\beta_2 + \gamma_{j[i]})\text{LogGrLivArea}_i + \beta_1 \text{OverallQual}_i \\ + \beta_3 \text{GarageCars}_i + \beta_4 \text{TotalBsmtSF}_i + \beta_5 \text{YearSinceRemodel}_i\end{aligned}$$

The neighborhood-specific effects satisfy

$$\alpha_j \sim \mathcal{N}(0, \tau_\alpha^2), \qquad \gamma_j \sim \mathcal{N}(0, \tau_\gamma^2), \qquad j = 1, \ldots, J.$$

Equivalently,

$$\begin{pmatrix} \alpha_j \\ \gamma_j \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \Sigma \right),$$

where

$$\Sigma = \begin{pmatrix} \tau_\alpha^2 & \rho\tau_\alpha\tau_\gamma \\ \rho\tau_\alpha\tau_\gamma & \tau_\gamma^2 \end{pmatrix}.$$

The priors are

$$\beta_0 \sim \mathcal{N}(8.8, 0.05^2),$$

$$\beta_1 \sim \mathcal{N}(0.10, 0.03^2), \quad \beta_2 \sim \mathcal{N}(0.25, 0.08^2), \quad \beta_3 \sim \mathcal{N}(0.06, 0.03^2),$$

$$\beta_4 \sim \mathcal{N}(0.00010, 0.00003^2), \quad \beta_5 \sim \mathcal{N}(-0.003, 0.004^2),$$

$$\tau_\alpha \sim \text{half-Normal}(0, 0.12^2), \qquad \tau_\gamma \sim \text{half-Normal}(0, 0.05^2),$$
$$\rho \sim \text{Uniform}(-1, 1), \qquad \sigma \sim \text{half-Normal}(0, 0.10^2).$$

This model is designed to capture two realistic features of the data: neighborhoods may have different baseline price levels, and the effect of living area may vary across neighborhoods. Since the project instructions allow the second model to be described methodologically without full fitting, we treat this model as the principled hierarchical extension of Model 1.

The Stan implementation of Model 2 is provided in the Appendix.

6 Posterior computation methods

For Model 1, we computed the posterior using two different methods:

1. Hamiltonian Monte Carlo (HMC) via Stan MCMC,
2. mean-field variational inference (VI) via Stan's variational approximation.

This comparison is important because the project requires two posterior computation methods, and it allows us to evaluate both approximation quality and computational reliability on the same model.

6.1 MCMC

We ran four MCMC chains with 1000 warmup iterations and 1000 sampling iterations per chain, using `adapt_delta = 0.95` and `max_treedepth = 12`. The chains finished successfully.

Posterior means from MCMC were:

$$\hat{\beta}_0 = 8.766, \quad \hat{\beta}_1 = 0.111, \quad \hat{\beta}_2 = 0.324, \quad \hat{\beta}_3 = 0.086,$$
$$\hat{\beta}_4 = 1.39 \times 10^{-4}, \quad \hat{\beta}_5 = -0.00332, \quad \hat{\sigma} = 0.171.$$

All reported $\hat{R}$ values were 1.00, and the effective sample sizes were large, indicating stable convergence and good mixing.

6.2 Variational inference

We also fit the same model using variational inference with 2000 output samples. The resulting approximation was poor. In particular, the estimated posterior mean for σ was approximately 3.90, which is clearly inconsistent with the MCMC posterior estimate of approximately 0.171. The variatoional approximation also produced substantially smaller posterior standard deviations compared to MCMC, suggesting underestemation of uncertainty. Moreover, the VI posterior predictive distribution had an implausibly wide range and produced extremely poor predictive summaries.

7 Diagnostics and posterior predictive checks

The MCMC trace plots showed no visible pathologies and no meaningful evidence of non-convergence. The posterior predictive checks were also reasonable: posterior predictive means tracked the observed log prices closely, and the posterior predictive density overlays matched the observed density well.

For MCMC, the predictive error metrics were

$$\mathrm{RMSE} = 0.1706, \qquad \mathrm{MAE} = 0.1200.$$

These values are consistent with a well-calibrated fit on the log-price scale.

By contrast, VI performed extremely poorly:

$$\mathrm{RMSE}_{\mathrm{VI}} = 5.464, \qquad \mathrm{MAE}_{\mathrm{VI}} = 5.069.$$

The VI posterior predictive means were badly misaligned with the observed response, and the predictive draws had an unrealistic range from about -28 to 27, far outside the observed log-price range.

To assess code correctness and posterior reliability, we used several validation strategies. First, prior predictive checks were used to verify that the prior choices implied plausible values on the log-price scale before fitting the model. Second, posterior predictive checks were used to confirm that the fitted model could reproduce the observed data distribution. Third, convergence diagnostics such as trace plots, $\hat{R}$, and effective sample sizes were used to evaluate the MCMC fit. Finally, the comparison between MCMC and variational inference served as an additional validation exercise: the severe mismatch in the VI posterior and predictive behavior highlighted that not all posterior approximations were trustworthy for this problem.

8 Discussion

Our results show that the naive Bayesian regression works well when fitted with MCMC. It captures the main patterns in the data, and the fitted model gives reasonable posterior summaries and posterior predictive checks.

The comparison between MCMC and VI is also very useful. Although VI is faster in principle, it did not approximate the posterior well in this project. In particular, the estimated residual scale under VI was far too large, and the posterior predictive results were clearly unrealistic. This suggests that MCMC is the more reliable method for this model.

Model 2 gives a natural extension of the baseline model. Even though we did not fully fit it, it improves the modeling framework by introducing neighborhood-level partial pooling. This allows both the baseline price level and the effect of living area to vary across neighborhoods, which is more consistent with the structure of the housing market.

9 Limitations

Our analysis has several limitations. First, the baseline model uses only a reduced subset of predictors, so some potentially useful information may have been left out. Second, the more complex hierarchical model was specified but not fully fitted, so its practical performance was not directly compared with Model 1. Third, the variational approximation used here appears to be particularly unreliable for this model, so we cannot treat it as a valid substitute for MCMC in this application.

10 Conclusion

In this project, we built a Bayesian regression model for log house prices and compared two posterior computation methods. MCMC gave stable inference and reasonable posterior predictive results, while VI performed poorly in both parameter estimation and prediction. We also proposed a hierarchical extension that allows neighborhood-level variation to enter the model more directly. Overall, this project shows that Bayesian regression can work well for structured real data, but posterior approximations still need to be checked carefully.

Team contributions

Ruohan Sun contributed approximately 45% of the project effort and was primarily responsible for the development of Model 2. Ruohan also took the lead in analyzing the model comparison results and writing the final report.

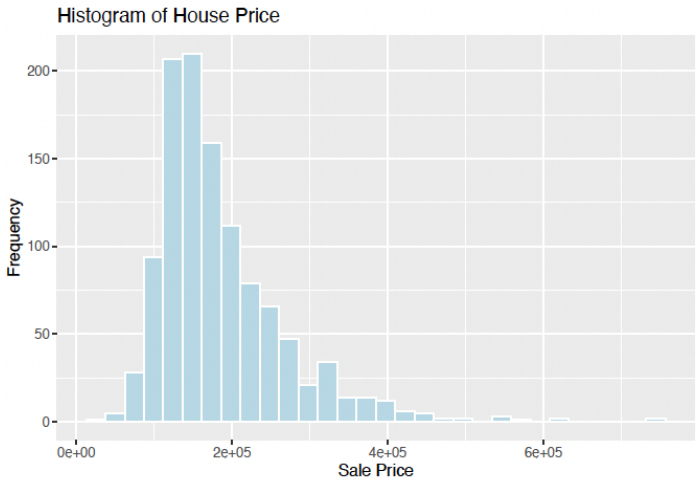
Haoyu You contributed approximately 55% of the project effort and was primarily responsible for the exploratory data analysis (EDA) and the implementation of Model 1. Haoyu also contributed to the preparation of figures and intermediate results used in the final report.

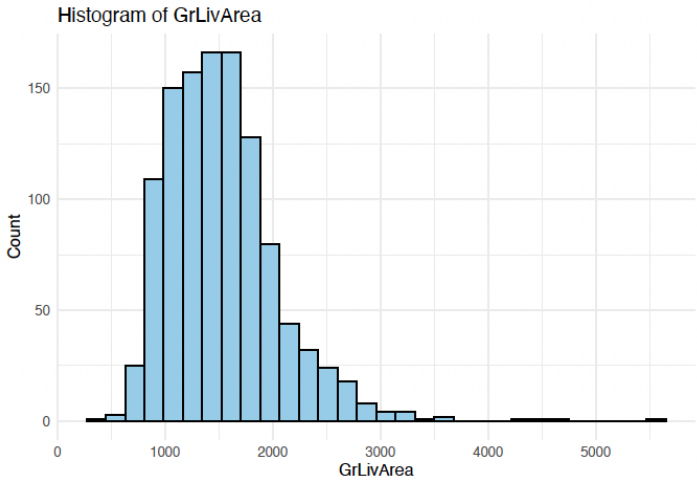
References

De Cock, D. (2011). Ames, Iowa: Alternative to the Boston housing data as an end-of-semester regression project. *Journal of Statistics Education, 19*(3).
Francke, M. K., Vos, G. A., & Janssen, R. J. A. (2004). The hierarchical trend model for property valuation and local price indices. *Journal of Real Estate Finance and Economics, 28*(2–3), 179–208.
Stan Development Team. (2024). *CmdStan user's guide: Variational inference.*
Wheeler, D. C., & Calder, C. A. (2014). A Bayesian approach to hedonic price analysis. *Journal of Housing Economics, 26*, 41–51.

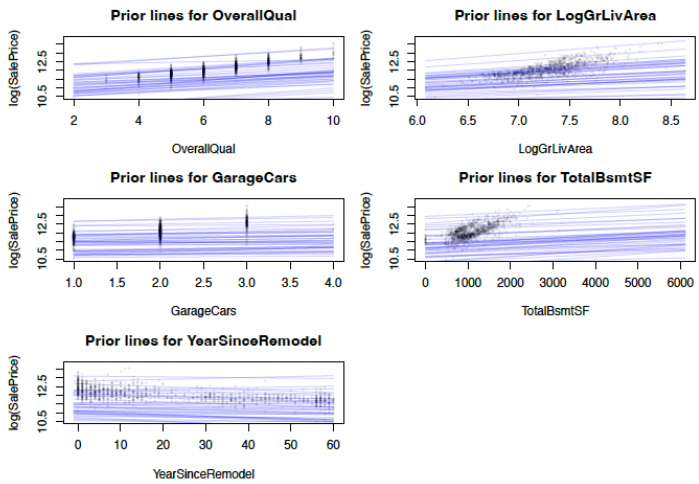
Appendix

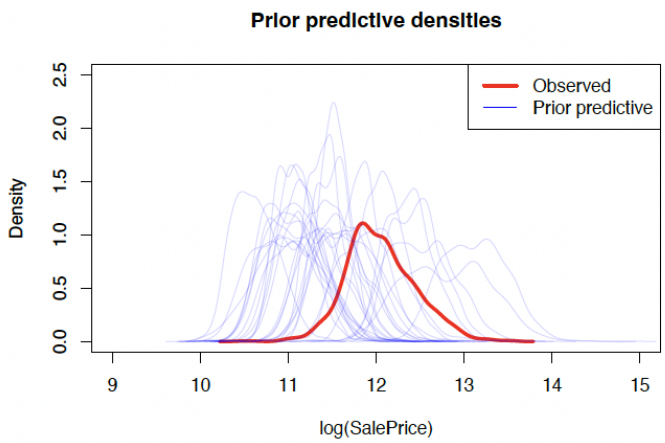
EDA



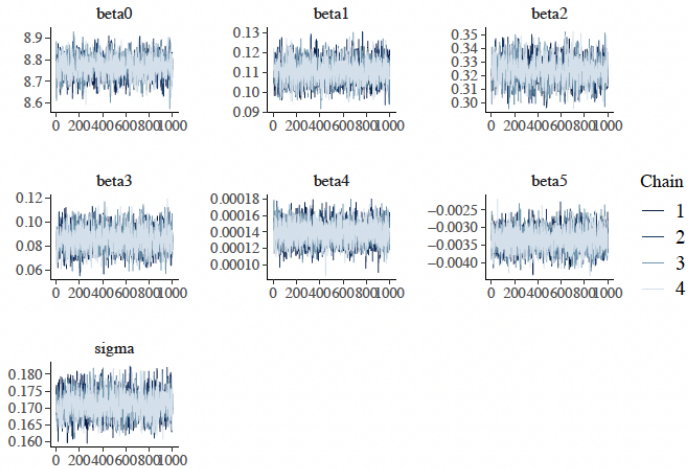


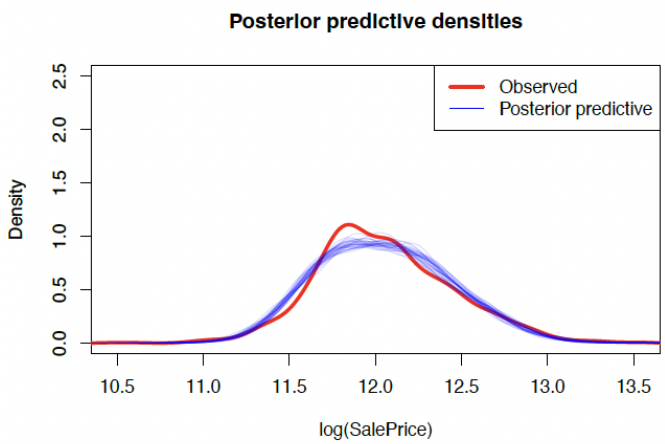
Prior Check


Prior lines for OverallQual
log(SalePrice)
OverallQual
Prior lines for LogGrLivArea
LogGrLivArea
Prior lines for GarageCars
GarageCars
Prior lines for TotalBsmtSF
TotalBsmtSF
Prior lines for YearSinceRemodel
YearSinceRemodel

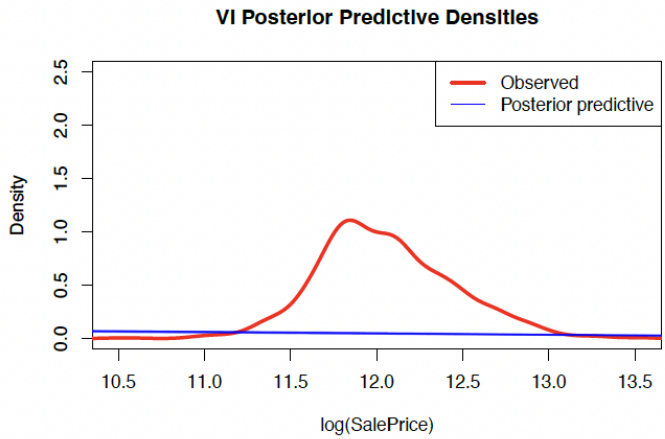

Prior predictive densities
Observed
Prior predictive
Density
log(SalePrice)

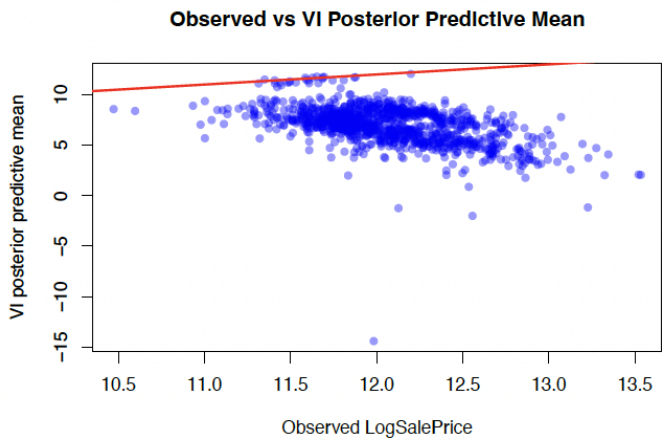
MCMC


beta0
beta1
beta2
beta3
beta4
beta5
Chain
1
2
3
4
sigma


Posterior predictive densities
Observed
Posterior predictive
Density
log(SalePrice)

VI


VI Posterior Predictive Densities
Observed
Posterior predictive
Density
0.0
0.5
1.0
1.5
2.0
2.5
10.5
11.0
11.5
12.0
12.5
13.0
13.5
log(SalePrice)


Observed vs VI Posterior Predictive Mean
VI posterior predictive mean
−15
−10
−5
0
5
10
10.5
11.0
11.5
12.0
12.5
13.0
13.5
Observed LogSalePrice

Comparison

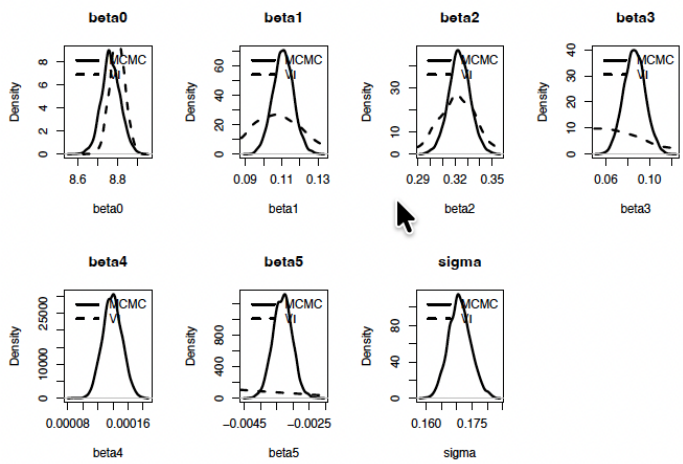


Table 1: Comparison of MCMC and VI estimates

Parameter	mean_mcmc	mean_vi	sd_mcmc	sd_vi	q5_mcmc	q5_vi	q95_mcmc	q95_vi
beta0	8.7665	8.7984	0.0477	0.0380	8.6908	8.7371	8.8464	8.8617
beta1	0.1108	0.1070	0.0056	0.0145	0.1014	0.0837	0.1200	0.1313
beta2	0.3235	0.3215	0.0087	0.0150	0.3091	0.2965	0.3378	0.3457
beta3	0.0862	0.0533	0.0097	0.0396	0.0704	-0.0085	0.1023	0.1222
beta4	0.000139	-0.004424	0.000013	0.000087	0.000118	-0.004564	0.000160	-0.004277
beta5	-0.003324	-0.006642	0.000299	0.003004	-0.003816	-0.01160	-0.002851	-0.001577
sigma	0.1709	3.9022	0.0036	0.0381	0.1650	3.8413	0.1772	3.9674

Stan code for Model 1

```
data {
  int<lower=1> N;

  vector[N] y;
  vector[N] OverallQual;
  vector[N] LogGrLivArea;
  vector[N] GarageCars;
  vector[N] TotalBsmtSF;
  vector[N] YearSinceRemodel;
}

parameters {
  real beta0;
  real beta1;
  real beta2;
  real beta3;
```

```
  real beta4;
  real beta5;
  real<lower=0> sigma;
}

model {
  vector[N] mu;

  beta0 ~ normal(8.8, 0.05);
  beta1 ~ normal(0.10, 0.03);
  beta2 ~ normal(0.25, 0.08);
  beta3 ~ normal(0.06, 0.03);
  beta4 ~ normal(0.00010, 0.00003);
  beta5 ~ normal(-0.003, 0.004);
  sigma ~ normal(0, 0.10);

  mu = beta0
      + beta1 * OverallQual
      + beta2 * LogGrLivArea
      + beta3 * GarageCars
      + beta4 * TotalBsmtSF
      + beta5 * YearSinceRemodel;

  y ~ normal(mu, sigma);
}

generated quantities {
  vector[N] mu;
  vector[N] y_rep;
  vector[N] log_lik;

  mu = beta0
      + beta1 * OverallQual
      + beta2 * LogGrLivArea
      + beta3 * GarageCars
      + beta4 * TotalBsmtSF
      + beta5 * YearSinceRemodel;

  for (n in 1:N) {
    y_rep[n] = normal_rng(mu[n], sigma);
    log_lik[n] = normal_lpdf(y[n] | mu[n], sigma);
  }
}
```

Stan code for Model 2

```
data {
  int<lower=1> N;
```

```
  int<lower=1> J;
  array[N] int<lower=1, upper=J> neighborhood;

  vector[N] y;
  vector[N] OverallQual;
  vector[N] LogGrLivArea;
  vector[N] GarageCars;
  vector[N] TotalBsmtSF;
  vector[N] YearSinceRemodel;
}

parameters {
  real beta0;
  real beta1;
  real beta2;
  real beta3;
  real beta4;
  real beta5;

  real<lower=0> sigma;

  vector<lower=0>[2] tau;

  cholesky_factor_corr[2] L_Omega;

  matrix[2, J] z_u;
}

transformed parameters {
  matrix[2, J] u;
  vector[J] alpha;
  vector[J] gamma;
  vector[N] mu;

  u = diag_pre_multiply(tau, L_Omega) * z_u;

  alpha = to_vector(u[1]');
  gamma = to_vector(u[2]');

  for (i in 1:N) {
    mu[i] =
      beta0
      + alpha[neighborhood[i]]
      + (beta2 + gamma[neighborhood[i]]) * LogGrLivArea[i]
      + beta1 * OverallQual[i]
      + beta3 * GarageCars[i]
      + beta4 * TotalBsmtSF[i]
      + beta5 * YearSinceRemodel[i];
  }
}
```

```
model {
  beta0 ~ normal(8.8, 0.05);
  beta1 ~ normal(0.10, 0.03);
  beta2 ~ normal(0.25, 0.08);
  beta3 ~ normal(0.06, 0.03);
  beta4 ~ normal(0.00010, 0.00003);
  beta5 ~ normal(-0.003, 0.004);

  tau[1] ~ normal(0, 0.12);
  tau[2] ~ normal(0, 0.05);

  sigma ~ normal(0, 0.10);

  L_Omega ~ lkj_corr_cholesky(2.0);

  to_vector(z_u) ~ normal(0, 1);

  y ~ normal(mu, sigma);
}

generated quantities {
  corr_matrix[2] Omega;
  matrix[2, 2] Sigma;
  vector[N] y_rep;

  Omega = multiply_lower_tri_self_transpose(L_Omega);
  Sigma = quad_form_diag(Omega, tau);

  for (i in 1:N) {
    y_rep[i] = normal_rng(mu[i], sigma);
  }
}
```